MAP EXTRACTION – THE WEB PAGE

Technical map in DXF for AutoCAD, GeoPackage and QGIS project, from the civil protection Drive, with nothing to install · English translation: in case of doubt the Italian text prevails · © 2026 Claude René Monbaron – UCS PC ANA PN e PC RFVG

Address

https://claudemonbaron.github.io/Carica_Mappe/estrazione/

Opens in any recent browser; designed for a **computer** (Chrome or Edge work best). The sister page for Garmin GPS units is https://claudemonbaron.github.io/Carica_Mappe/ (link at the bottom of each page).

What it does

1. For those who cannot or do not want to install the app. On old computers or with little memory (4 GB) it is fine for small municipalities; for large ones the app is better.
2. It does the same extraction as the **Estrarre_Mappe** program for Mac and Windows, with the same program running in the browser (Python): the files are identical to those of the app.
3. It downloads from the Drive only the sheets it needs, cuts them to the municipality (or rectangle), joins them into one DXF with the **500 m grid** and prepares the chosen versions.
4. Regions and countries: all Italian regions, Switzerland, Slovenia and Austria (RDN2008 / ETRS89 UTM zone 32 or 33; municipalities in different zones: everything in UTM33).
5. The first opening loads the program (about 30 MB); afterwards it starts in a few seconds. Downloaded sheets stay in the browser.
6. **Svuota la cache** (empty the cache): the page button deletes the sheets and tiles stored in the browser (after a confirmation). Use it if the browser reports little space or an extraction gives strange errors; next time the page downloads again only what it needs.

1 · Municipality and additional municipalities

1. Choose **Region / Country**, **Province** (Land or Canton abroad) and type the **Municipality**.
2. **Only the municipality**, or **Neighbours** (also of other regions or countries), or **Rectangle**: the km beyond the municipality towards North, South, East and West (5 by default) and **Find municipalities**; the map is cut exactly to the rectangle.

2 · What to prepare

1. **AutoCAD**: DXF in UTM metres, in colour and in black and white for printing (with .prj files).
2. **DXF in WGS84 degrees**: for Global Mapper and GPS.
3. **GeoPackage** UTM and WGS84: for QGIS and QField on phones and tablets.
4. **QGIS project** (`.qgs`), with the orthophotos of the country (internet needed).
5. **Terrain and 3D views in QGIS** (unticked by default): the project opens with the terrain of the area and two ready-made 3D views. Downloads a few MB more.
6. **UTM terrain for CAD** (unticked by default): the terrain in metres, in the zone of the DXF files (UTM32 or UTM33), for Civil 3D or similar programs.

3 · Extract and save

1. Before starting, the page shows how many **municipalities**, how many **files** (DXF and terrain tiles) and how many **MB** it will download.
2. **Chrome or Edge on a computer**: **Choose where to save** (for example Downloads) and allow Chrome to edit the files. **Safari, Firefox, phones and tablets**: the result arrives as a `.zip` file in Downloads.

3. **Estrai** (Extract): keep the page open until the green "Fatto" (Done) appears. For very large extractions (over 100 MB) the app is better.

Opening the result

1. **AutoCAD**: double-click the DXF; the `reticolo` layer is the red 500 m grid. **QGIS** (free, https://qgis.org/download/, version 4.4 or later): open the `.qgs` file.
2. **3D in QGIS** (with the terrain box): **View › 3D Map Views › 3D dall'alto** (from above) or **3D in prospettiva** (perspective). The 3D view does not drag the 2D map; the area seen in 3D is outlined on the 2D map.
3. **3D prints and buttons** (with the terrain box): Project › Layouts also contains four 3D layouts (`A4_Vert_3D`, `A4_Oriz_3D`, `A3_Vert_3D`, `A3_Oriz_3D`). When the project opens, QGIS warns that it contains macros: click **Enable macros**. Three buttons then appear in the toolbar: **3D sulla vista 2D** (3D on the 2D view) moves the 3D view to the area shown in the 2D map; **2D sulla vista 3D** (2D on the 3D view) does the opposite; **Stampe 3D come la vista 3D** (3D prints like the 3D view) copies the framing of the open 3D view into the four 3D layouts. Open the 3D view first; save the project to keep the framing. Without enabling macros the project still works, but without the three buttons.
4. **Printing in QGIS**: Project › Layouts contains eight ready-made layouts, A4 and A3, portrait and landscape, in WGS84 and in the UTM system of the DXF files (UTM32 or UTM33), with grid and coordinates, scale bar, north arrow and the municipality name as title. The scale is locked at 1:5000: to frame another area, move the map content (C key) or use “Set Map Extent to Match Main Canvas Extent”: the area changes, the scale stays. In the WGS84 layouts the map is drawn as in the UTM ones (true 1:5000 scale, same area); the grid and coordinates are in WGS84 degrees and the arrow points to true north.
5. **Global Mapper**: open `<municipality>.gmw` (terrain with hillshade and colour DXF; black-and-white DXF, GeoPackage and Geoportale orthophotos switched off, to be switched on in the Control Center) or `<municipality>_3D.gmw`, which also opens the 3D view. Files are looked for in the folder of the `.gmw`, so the extraction folder can be moved.
6. Sources, licences and **disclaimer**: see the PDF in the `_____ISTRUZIONI_DISCLAIMER` folder of Estrazione_mappe. The page has no accounts and no cookies; visits are counted anonymously (GoatCounter).

INSTRUCTIONS FOR USE – MAP EXTRACTION (DXF, QGIS, GEOPACKAGE)

Estrarre_Mappe program, with disclaimer · English translation: in case of doubt the Italian text prevails · © 2026 Claude René Monbaron – UCS PC ANA PN e PC RFVG

a) Contents of the Estrazione_mappe folder

1. `____Estrazione_Mappe.dmg` (Mac with Apple chip M1–M4) and `____Estrazione_Mappe.exe` (Windows): the program. It is the only thing to install.
2. Folders `<region>_dxf_...`: the technical map in DXF, one zip file per sheet or per municipality. The program takes only the ones it needs: there is no need to open them by hand.
3. `_dati`: lists of municipalities, sheets and neighbours, QGIS project and 3D templates, list of orthophotos. The program reads them by itself.
4. `DTM_TESSERE`: the terrain (10 m digital terrain model) in tiles, in WGS84 degrees and in UTM32 and UTM33 metres. Used for 3D in QGIS and for terrain in CAD; the program takes only the tiles of the area.
5. `Qgis_ONLINE`: ready-made QGIS projects with the maps and orthophotos available online, one per region and one with all regions.
6. Regions and countries available, with the coordinate system of the DXF files (RDN2008 / ETRS89, UTM zone 32 or 33): Friuli Venezia Giulia (UTM33), Veneto (UTM32 and UTM33), Trentino, Alto Adige/South Tyrol, Aosta Valley, Piedmont, Lombardy, Liguria, Emilia-Romagna, Tuscany, Sardinia (UTM32), Marche, Umbria, Lazio, Abruzzo, Molise, Campania, Apulia, Basilicata, Calabria, Sicily (UTM33), Switzerland (UTM32), Slovenia and Austria (UTM33). When municipalities in different zones are extracted together, everything is converted to UTM33.

b) Installing the program

1. **Mac**: double-click the `.dmg` file, drag `____Estrazione_Mappe` to the Desktop or to Applications, then eject the dmg. The first time, open it with right-click › Open (the program is not signed by Apple). To update: move the old version to the Bin and repeat.
2. **Windows**: copy the `.exe` file anywhere (for example to the Desktop) and double-click it. If the Windows SmartScreen warning appears: More info › Run anyway.
3. The version number is in the window title. Settings (folders, Drive link) are kept after an update.

c) Extracting a municipality

1. **Data from**: choose **Shared Drive** (the link is already filled in). The program downloads only the sheets it needs and keeps them in a folder on the computer (`Estrarre_Mappe_dati_Drive`, in the user's home folder): the second time they are not downloaded again, and sheets updated on the Drive are downloaded again automatically. At every start the lists are updated; without a network the program works with the lists already downloaded.
2. **Svuota cache** (empty cache, with «Shared Drive»): after a confirmation it deletes the sheets and terrain tiles already downloaded in the `Estrarre_Mappe_dati_Drive` folder; the lists are kept. Useful when disk space is short or an extraction gives strange errors: next time the program downloads again only what it needs.
3. **Work folder**: where the extractions are created (for example Documents).
4. Choose region, province and municipality (the name can be typed to search it), then any **neighbouring** municipalities, also in other regions or countries, and, if needed, only some sheets.
5. **Rectangle** (instead of the neighbours): choose "Rettangolo" and enter how many km beyond the municipality towards North, South, East and West (5 km by default). The program finds all the municipalities touching the rectangle and the map is cut exactly to it.

6. **What to prepare**: AutoCAD (DXF in UTM metres, in colour and in black and white for printing), DXF in WGS84 degrees, GeoPackage, QGIS project. Two more boxes, unticked by default because they download more data (a few MB per municipality):

- **Terrain and 3D views in QGIS**: the QGIS project opens with the terrain of the area and two ready-made 3D views (see d).

- **UTM terrain for CAD (Civil 3D)**: the terrain in metres, in the same zone as the DXF files (UTM32 or UTM33), to build a surface in Civil 3D or similar programs; plain AutoCAD does not use it.

7. **Estrai** (Extract). The log at the bottom shows the progress; at the end the button opens the folder `<Municipality>+` with the files. With many neighbours the program may make one DXF per municipality, plus the file `CARICA_TUTTI_AutoCAD.lsp` to load them together.

d) Opening the result

1. **AutoCAD**: double-click the DXF. For several municipalities: open one and drag `CARICA_TUTTI_AutoCAD.lsp` into the window; it inserts the others at their coordinates. The `reticolo` layer is the red 500 m grid.
2. **QGIS**, free for Windows, Mac and Linux: https://qgis.org/download/ (version 4.4 or later is recommended: 3.x versions open these projects badly, and on Mac with macOS 27.2 version 4.2 does not start). Open the `.qgs` file of the extraction: themes (buildings, roads, water, vegetation, contours...) are switched on and off in the Layers panel.
3. **3D in QGIS** (with the "Terrain and 3D views" box): the project contains the `terreno_WGS84` layer, switched off in the 2D map, and two views already set on the area: **View › 3D Map Views › 3D dall'alto** (from above) or **3D in prospettiva** (perspective). Map, orthophotos and OpenStreetMap are draped on the relief. The 3D view does not drag the 2D map (so QGIS stays fast); the area seen in 3D is outlined on the 2D map. The tiles are in the `terreno` subfolder, with the file `terreno_WGS84.vrt` (and `terreno_UTM32.vrt` or `terreno_UTM33.vrt` if requested). The terrain also comes in two styles for the 2D map: `(colori)`, semi-transparent elevation bands, and `(b/n)`, hillshade suited to black-and-white printing.
4. **3D prints and buttons** (with the terrain box): Project › Layouts also contains four 3D layouts (`A4_Vert_3D`, `A4_Oriz_3D`, `A3_Vert_3D`, `A3_Oriz_3D`). When the project opens, QGIS warns that it contains macros: click **Enable macros**. Three buttons then appear in the toolbar: **3D sulla vista 2D** (3D on the 2D view) moves the 3D view to the area shown in the 2D map; **2D sulla vista 3D** (2D on the 3D view) does the opposite; **Stampe 3D come la vista 3D** (3D prints like the 3D view) copies the framing of the open 3D view into the four 3D layouts. Open the 3D view first; save the project to keep the framing. Without enabling macros the project still works, but without the three buttons.
5. **Orthophotos** in the QGIS project (internet needed): Italy from the Geoportale Nazionale, Switzerland SWISSIMAGE, Austria basemap.at, Slovenia GURS. They are added automatically according to the country of the municipalities.
6. **GeoPackage** (`.gpkg`): opens in QGIS, in QField on phones and tablets and in other GIS programs.
7. **Printing in QGIS**: Project › Layouts contains eight ready-made layouts, A4 and A3, portrait and landscape, in WGS84 and in the UTM system of the DXF files (UTM32 or UTM33), with grid and coordinates, scale bar, north arrow and the municipality name as title. The scale is locked at 1:5000: to frame another area, move the map content (C key) or use “Set Map Extent to Match Main Canvas Extent”: the area changes, the scale stays. In the WGS84 layouts the map is drawn as in the UTM ones (true 1:5000 scale, same area); the grid and coordinates are in WGS84 degrees and the arrow points to true north.
8. **Global Mapper**: open `<municipality>.gmw` (terrain with hillshade and colour DXF; black-and-white DXF, GeoPackage and Geoportale orthophotos switched off, to be switched on in the Control Center) or `<municipality>_3D.gmw`, which also opens the 3D view. Files are looked for in the folder of the `.gmw`, so the extraction folder can be moved.

e) Without AutoCAD: free programs to view and print to scale

1. **QGIS** (recommended): the project already contains print layouts at 1:5000 (see d); for another scale: Project › New Print Layout › Add Map, set the scale (for example 1:5000 or 1:10000) and export to PDF or print. Orthophotos and GPX tracks of the assigned areas can be overlaid.
2. **Autodesk DWG TrueView** (free, Windows only): https://www.autodesk.com/products/dwg-trueview/overview – opens and prints DXF and DWG to scale, without editing them.
3. **LibreCAD** (free, Windows, Mac, Linux): https://librecad.org – becomes slow with very large DXF files: in that case QGIS is better.

f) Limits

1. The data are those published by the authorities, with their survey dates (from 1979 to 2025): recent buildings or roads may be missing.
2. Small discrepancies may remain at the borders between regions and countries: each authority maps up to its own border with its own specifications.

g) Disclaimer

These extractions are taken directly from the official technical maps and databases published by the authorities listed in the Italian text (point h), downloaded between August and October 2026, and transformed only as needed (joining the sheets, coordinate system, layer names, grid). Use of the program and of the data is free of charge. The undersigned makes them available to all civil protection operators, volunteers or not, and declines any responsibility for inaccuracies and for improper use of these data. Suggestions and reports: claude.monbaron59@gmail.com. **Warning:** the Abruzzo data are licensed CC BY-NC: maps and files containing them may not be used for commercial purposes. The licences of the original data remain those of the respective authorities (full list of sources in the Italian text, point h).

h) Fonti dei dati e licenze

Le licenze dei dati originali restano quelle dei rispettivi enti; dove indicata, la licenza CC BY richiede di citare la fonte, come si fa qui.

1. **Friuli Venezia Giulia** – Carta Tecnica Regionale Numerica CTRN 1:5000 in DXF, Regione Autonoma Friuli Venezia Giulia (IRDAT / EagleFVG).
2. **Veneto** – Carta Tecnica Regionale 1:5000, Regione del Veneto (geoportale IDT).
3. **Trentino** – Carta Tecnica Provinciale, Provincia autonoma di Trento.
4. **Alto Adige** – geodati della Provincia autonoma di Bolzano.
5. **Valle d'Aosta** – Carta Tecnica Regionale 1:5000 (CC0) e 1:10.000; Repertorio cartografico SCT della Regione Autonoma Valle d'Aosta (CC BY 4.0).
6. **Piemonte** – Base Dati Territoriale di Riferimento degli Enti BDTRE, Regione Piemonte (CC BY 4.0).
7. **Lombardia** – Database Geotopografico DBGT, Regione Lombardia (CC BY 4.0).
8. **Liguria** – Carta Tecnica Regionale / DB Topografico, Regione Liguria.
9. **Emilia-Romagna** – Database Topografico Regionale DBTR (classi a libero accesso), DTM 5x5 m, uso del suolo 2023, Regione Emilia-Romagna (CC BY 4.0).
10. **Toscana** – Carta Tecnica Regionale 1:10.000 / DB Topografico, Regione Toscana (Geoscopio, CC BY 4.0); isole minori (Gorgona, Capraia, Montecristo, Giglio, Giannutri) da OpenStreetMap con curve di livello ricavate da TINITALY 1.1 (INGV, CC BY 4.0).
11. **Marche** – Database Geotopografico DBGT 2019, Regione Marche (CC BY 4.0).
12. **Umbria** – Carta Tecnica Regionale numerica 1:10.000 e 1:5000, Regione Umbria (CC BY).
13. **Lazio** – Database Geotopografico DBGT 1:5000 edizione 2014, Regione Lazio (CC BY 4.0).
14. **Campania** – Carta Tecnica Numerica e Data Base Topografico 1:5000 aggiornati al 2020, Regione Campania (CC BY 4.0).
15. **Abruzzo** – Database Territoriale Regionale D.B.T.R. 1:5000 edizione 2007, Regione Abruzzo (CC BY-NC: **solo uso non commerciale**).
16. **Puglia** – Carta Tecnica Regionale 1:5000 (volo 2006), Regione Puglia, SIT Puglia (IODL 2.0).
17. **Svizzera** – swissTLM3D, swissNAMES3D, swissBOUNDARIES3D e swissALTI3D, Ufficio federale di topografia swisstopo (dati aperti, © swisstopo).
18. **Slovenia** – modello topografico DTM 1:5000, nomi geografici REZI, registro delle unità territoriali RPE, Geodetska uprava Republike Slovenije GURS (CC BY 4.0).
19. **Austria** – Digitales Landschaftsmodell DLM, curve di livello DGM Höhenschichtlinien, confini dei comuni VGD, Bundesamt für Eich- und Vermessungswesen BEV (CC BY 4.0); copertura del suolo da OpenStreetMap.
20. **Molise** – da OpenStreetMap (ODbL), con curve di livello ricavate da TINITALY 1.1 (INGV, CC BY 4.0), in attesa dei dati regionali.
21. **Basilicata** – Database Geotopografico DBGT 2013, Regione Basilicata, RSDI (CC BY 4.0).
22. **Calabria** – Database Geotopografico DBGT 1:5000 2007–2008, Regione Calabria (dati aperti).
23. **Sicilia** – Carta Tecnica Regionale 1:10.000 2012–2013, Regione Siciliana, SITR (CC BY 4.0); boschi integrati da OpenStreetMap (ODbL).
24. **Sardegna** – Database Geotopografico 1:10.000 (DBGT_10K_22), Regione Autonoma della Sardegna.
25. **Terreno (DTM) per il 3D** – TINITALY 1.1, INGV (Tarquini et al. 2007 e 2012, CC BY 4.0); in Svizzera swissALTI3D (© swisstopo); in Austria DGM, Datenquelle BEV (CC BY 4.0); in Slovenia DMV, GURS (CC BY 4.0).
26. **Confini dei comuni italiani** – ISTAT, Limiti amministrativi al 1° gennaio 2026 (CC BY).
27. **OpenStreetMap** – © OpenStreetMap contributors, licenza ODbL (openstreetmap.org/copyright).

28. **Ortofoto online** – Geoportale Nazionale (Ministero dell'Ambiente e della Sicurezza Energetica); SWISSIMAGE © swisstopo; basemap.at (CC BY 4.0); ortofoto GURS (CC BY 4.0).

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